



WEB DESIGN & DEVELOPMENT

ASSIGNMENT



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ASSIGNMENT#1

FORM VALIDATION

HTML code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Form Validation</title>
5   <link rel="stylesheet" href="Form Validation.css">
6 </head>
7 <body>
8
9   <form id="form">
10     <label>First Name</label><br>
11     <input type="text" id="name"><br><br>
12     <label>Last Name</label><br>
13     <input type="text" id="lname"><br><br>
14     <label>Email</label><br>
15     <input type="email" id="email"><br><br>
16     <label>Contact Number</label><br>
17     <input type="text" id="contact"><br><br>
18     <label>Password</label><br>
19     <input type="password" id="password"><br><br>
20
21     <label>Confirm Password</label><br>
22     <input type="password" id="confirm"><br><br>
23
24     <button type="submit">Submit</button>
25   </form>
26
27   <script src="Form Validation.js"></script>
28 </body>
29 </html>
```

Code Details – Form Validation (HTML File)

Document Declaration

- The document begins with the HTML5 declaration which informs the browser about the HTML version being used.

HTML Root Element

- The html tag marks the beginning of the HTML document and contains the entire webpage content.

Head Section

- The head tag is used to store meta information about the webpage.
- The title tag sets the name of the webpage shown in the browser tab as “Form Validation”.

- The link tag connects an external CSS file named “Form Validation.css” which is used to style the form.
-

Body Section

- The body tag contains all the visible content of the webpage.
-

Form Element

- The form tag is used to create a form for user input.
 - The form is given an id named “form” so it can be accessed in JavaScript for validation.
-

First Name Label and Input

- The label tag displays the text “First Name” for user guidance.
 - The input field accepts text input from the user.
 - The input field is assigned an id “name” which helps JavaScript access the entered value.
 - Line breaks are used to move the input field to a new line.
-

Last Name Label and Input

- The label tag displays the text “Last Name”.
 - The text input field allows the user to enter their last name.
 - The id “lname” uniquely identifies this input field for validation.
-

Email Label and Input

- The label tag displays the text “Email”.
 - The email input field is used to enter an email address.
 - The id “email” is used to fetch the email value in JavaScript.
-

Contact Number Label and Input

- The label tag displays the text “Contact Number”.
 - The text input field allows the user to enter a contact number.
 - The id “contact” is used to validate the contact number using JavaScript.
-

Password Label and Input

- The label tag displays the text “Password”.
 - The password input field hides the characters entered by the user.
 - The id “password” is used to access the password value.
-

Confirm Password Label and Input

- The label tag displays the text “Confirm Password”.
 - The password input field is used to re-enter the password.
 - The id “confirm” is used to check whether both passwords match.
-

Submit Button

- The button element is used to submit the form.
 - The button type is set to submit so that form submission can be triggered.
-

JavaScript File Linking

- The script tag is used to connect the external JavaScript file named “Form Validation.js”.
 - JavaScript handles the validation logic of the form.
-

Closing Tags

- The form tag is closed after all input fields and the submit button.
 - The body and html tags are closed to properly end the document.
-

CSS code:

```
1  body {
2      font-family: Arial, sans-serif;
3      background-color: #1e1e2f;
4      display: flex;
5      justify-content: center;
6      align-items: center;
7      height: 100px;
8  }
9  /* Form box */
10 form {
11     background-color: #2b2b3d;
12     padding: 25px;
13     width: 320px;
14     border-radius: 8px;
15 }
16 /* Labels */
17 label {
18     color: #cfcfe6;
19     font-size: 14px;
20 }
21 /* Input fields */
22 input {
23     width: 90%;
24     padding: 9px;
25     margin-top: 5px;
26     margin-bottom: 15px;
27     border: 1px solid #444;
28     border-radius: 5px;
29     background-color: #1e1e2f;
30     color: #ffffff;
31 }
```

```
32 /* Button */
33 button {
34     width: 100%;
35     padding: 10px;
36     background-color: #6c8cff;
37     color: white;
38     border: none;
39     border-radius: 5px;
40     font-size: 15px;
41     cursor: pointer;
42 }
43 /* Button hover */
44 button:hover {
45     background-color: #5a7be0;
46 }
```

CSS Code Details – Form Validation

Body Styling

- The **body** selector applies styles to the **entire webpage**.
- **font-family** sets the text style to **Arial** for better readability.
- **background-color** applies a **dark background** to the page.
- **display: flex** enables **flexbox layout**.
- **justify-content: center** aligns the form **horizontally** at the center.
- **align-items: center** aligns the form **vertically** at the center.
- **height** controls the vertical space used for centering.

Form Container Styling

- The **form** selector styles the **form box**.
- **background-color** gives the form a **slightly lighter dark shade**.

- **padding** creates **space inside** the form.
 - **width** sets a **fixed size** for the form layout.
 - **border-radius** adds **rounded corners** for a smooth look.
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Label Styling

- The **label** selector styles all form labels.
 - **color** sets a **light text color** for visibility on dark background.
 - **font-size** controls the **text size** of labels.
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Input Field Styling

- The **input** selector styles all input fields.
 - **width** defines the **input box size**.
 - **padding** adds **inner spacing** for comfortable typing.
 - **margin-top** and **margin-bottom** add **spacing between elements**.
 - **border** creates a **visible outline** around inputs.
 - **border-radius** gives **rounded edges**.
 - **background-color** matches the dark theme.
 - **color** sets the **input text color** to white.
-

Button Styling

- The **button** selector styles the submit button.
- **width: 100%** makes the button span the full form width.
- **padding** increases the **clickable area**.
- **background-color** applies a **soft blue accent color**.
- **color** sets the **button text color** to white.
- **border: none** removes default borders.
- **border-radius** smoothens button edges.

- **font-size** controls the button text size.
 - **cursor: pointer** changes the cursor to **hand symbol** on hover.
-

Button Hover Effect

- The **button:hover** selector applies style when the mouse is over the button.
 - **background-color** changes to a **darker shade** on hover.
 - This provides **visual feedback** to the user.
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Final Conclusion

- This CSS code improves **layout, alignment, and visual appearance**.
 - It uses **flexbox, dark theme, and hover effects** for a modern design.
 - The form becomes **user-friendly, readable, and interactive**.
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Java script code:

```
1 document.getElementById("form").addEventListener("submit", function(e) {
2     e.preventDefault();
3     let name = document.getElementById("name").value;
4     let lname = document.getElementById("lname").value;
5     let email = document.getElementById("email").value;
6     let contact = document.getElementById("contact").value;
7     let password = document.getElementById("password").value;
8     let confirm = document.getElementById("confirm").value;
9     if (name == "" || lname == "" || email == "" || password == "" || confirm == "") {
10         alert("All fields are required");
11         return;
12     }
13     if (email.indexOf("@") == -1) {
14         alert("Invalid email");
15         return;
16     }
17     if (contact != "") {
18         if (isNaN(contact)) {
19             alert("Contact number must contain only numbers");
20             return;
21         }
22         if (contact.length != 10) {
23             alert("Contact number must be 10 digits");
24             return;
25         }
26     }
```

```
27     if (password.length < 6) {
28         alert("Password must be at least 6 characters");
29         return;
30     }
31     if (password != confirm) {
32         alert("Passwords do not match");
33         return;
34     }
35     alert("Form submitted successfully");
36 });
```

JavaScript Code Details – Form Validation

Accessing the Form and Event Handling

- **document** refers to the entire HTML document.
- **getElementById("form")** selects the form using its **id**.

- **addEventListener("submit")** listens for the **submit button click**.
 - **function(e)** defines a function that runs when the form is submitted.
 - **e.preventDefault()** stops the form from submitting automatically.
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Reading User Input Values

- **let** is used to declare variables.
 - **document.getElementById("name").value** stores the **first name input**.
 - **document.getElementById("lname").value** stores the **last name input**.
 - **document.getElementById("email").value** stores the **email input**.
 - **document.getElementById("contact").value** stores the **contact number**.
 - **document.getElementById("password").value** stores the **password**.
 - **document.getElementById("confirm").value** stores the **confirm password**.
-

Empty Field Validation

- **if** condition checks whether any required field is empty.
 - **== ""** checks for empty input.
 - **|| (OR operator)** checks multiple conditions together.
 - **alert()** displays an error message.
 - **return** stops further execution if validation fails.
-

Email Validation

- **email.indexOf("@")** checks whether the **@ symbol** exists.
 - **== -1** means the symbol is not found.
 - **alert("Invalid email")** displays an email error message.
 - **return** stops form submission.
-

Optional Contact Number Validation

- **if (contact != "")** checks whether the contact field is filled.
 - **isNaN(contact)** checks if the value is **not a number**.
 - **alert()** shows an error if non-numeric characters are found.
 - **contact.length != 10** checks whether the number has exactly **10 digits**.
 - **return** stops execution if contact validation fails.
-

Password Length Validation

- **password.length** checks the number of characters in the password.
 - **< 6** ensures the password is at least **6 characters long**.
 - **alert()** displays a password length error.
-

Confirm Password Validation

- **password != confirm** checks whether both passwords match.
 - **alert("Passwords do not match")** displays an error message.
 - **return** prevents form submission.
-

Successful Submission

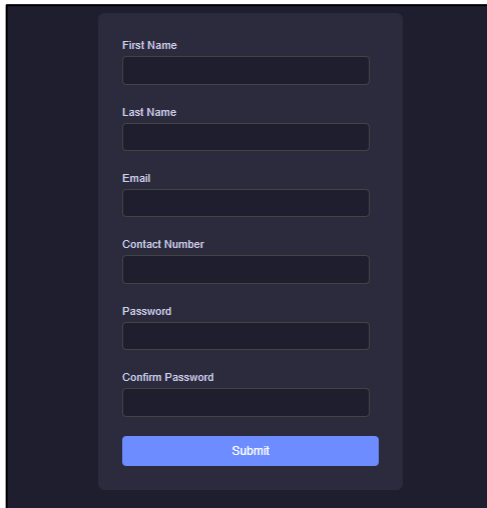
- **alert("Form submitted successfully")** is shown when all validations pass.
 - The form is considered valid after this message.
-

Final Conclusion

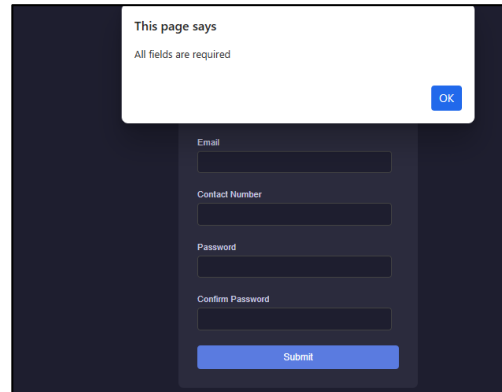
- This JavaScript code performs **client-side validation**.
 - It prevents incorrect data from being submitted.
 - It improves **user experience** and **data accuracy**.
 - The form is submitted only when **all validation conditions are satisfied**.
-

Output:

All fields are Mandatory:

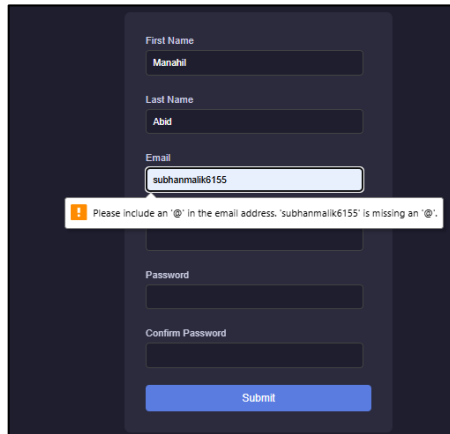


A registration form with the following fields: First Name, Last Name, Email, Contact Number, Password, and Confirm Password. All fields are empty. A blue 'Submit' button is at the bottom.

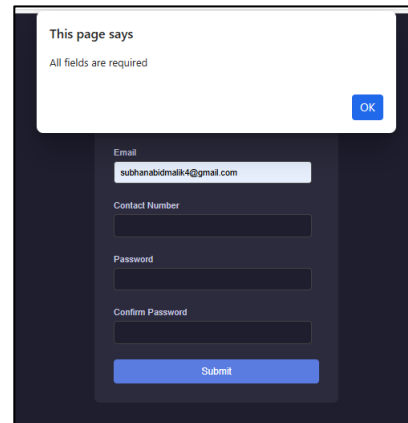


The same registration form as before, but with a white error message box at the top that says "This page says All fields are required". There is a blue 'OK' button next to the message. The form fields are still empty.

Email Validation:

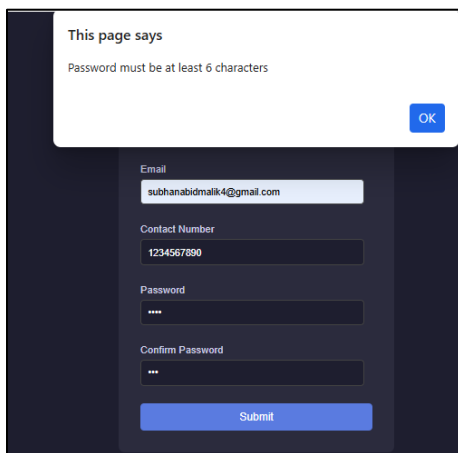


The registration form with the First Name field filled with "Manahil", Last Name with "Abid", and Email with "subhanmalik6155". A white error message box is displayed over the Email field, stating: "Please include an '@' in the email address. 'subhanmalik6155' is missing an '@'". The Password and Confirm Password fields are empty. A blue 'Submit' button is at the bottom.

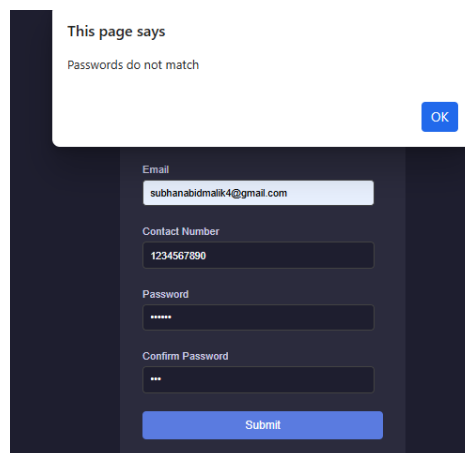


The registration form with the First Name field filled with "Manahil", Last Name with "Abid", and Email with "subhanabidmalik4@gmail.com". A white error message box at the top says "This page says All fields are required". There is a blue 'OK' button next to the message. The Contact Number, Password, and Confirm Password fields are empty.

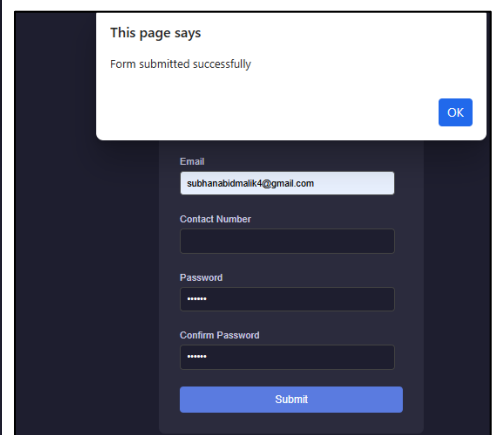
Password Validation:



The registration form with the First Name field filled with "Manahil", Last Name with "Abid", and Email with "subhanabidmalik4@gmail.com". A white error message box at the top says "This page says Password must be at least 6 characters". There is a blue 'OK' button next to the message. The Contact Number field is filled with "1234567890", the Password field is filled with "****", and the Confirm Password field is filled with "***".



The registration form with the First Name field filled with "Manahil", Last Name with "Abid", and Email with "subhanabidmalik4@gmail.com". A white error message box at the top says "This page says Passwords do not match". There is a blue 'OK' button next to the message. The Contact Number field is filled with "1234567890", the Password field is filled with "*****", and the Confirm Password field is filled with "***".



The registration form with the First Name field filled with "Manahil", Last Name with "Abid", and Email with "subhanabidmalik4@gmail.com". A white message box at the top says "This page says Form submitted successfully". There is a blue 'OK' button next to the message. The Contact Number field is filled with "1234567890", the Password field is filled with "*****", and the Confirm Password field is filled with "*****".

Not mandatory contact field validation:

This page says
Contact number must be 10 digits

OK

Email
subhanabidmalik4@gmail.com

Contact Number
12345

Password

Confirm Password

Submit

This page says
Contact number must contain only numbers

OK

Email
subhanabidmalik4@gmail.com

Contact Number
123456789q

Password

Confirm Password

Submit

This page says
Form submitted successfully

OK

Email
subhanabidmalik4@gmail.com

Contact Number
1234567891

Password

Confirm Password

Submit

✦ THE END ✦